INCH-POUND
MIL-DTL-24640/7D
22 August 2002
SUPERSEDING
MIL-C-24640/7C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-CONDUCTOR, AWG 22, TYPE 1XMSO

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, AWG size 22.
Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling. Component wire color shall be white, unless specified otherwise.
Third - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with WC-27500.
Fourth - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.
Fifth - The required number of shielded and jacketed singles shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.
Sixth - Binder tape applied helically with overlap.
Seventh - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Eighth - Binder tape applied helically with overlap, at manufacturer's option.
Ninth - Crosslinked polyolefin jacket. (See Table I for thickness).

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TABLE I. Construction details.

PIN M24640/7	Type number	Number of component wires	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	1XMSO-7	7	0.247	0.038	0.344	0.370	116.0
-02UO	1XMSO-16	16	.379	.041	.471	.507	212.0
-03UO	1XSMO-70	70	.785	.044	.859	.925	735.0

REQUIREMENTS:

Qualification required.
Construction details: See Table I.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Flammability: 30 seconds (max); 3.0 inch (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak) or 3 kHz spark test
5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

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EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.11
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	Not applicable
Conductor to shield	1000
Shield to shield	200
Component shield to overall shield	500
Insulation resistance - megohms-1000 feet, minimum	
Conductor to shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield)	100
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	120
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	13 min, 30 max
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
Cross link proof test (percent maximum)	
Jacket	50
Drip	Not applicable
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	4
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60

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Permanence of printing (conductor) - cycles, minimum ..... Not applicable
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage Not applicable
Heat distortion - percent of unaged, maximum 30
Cable sealant removability ..... Not applicable
Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance
Milliohms per meter, maximum 700
EMP response, dB, minimum 60
Pulse response time - ms/500 feet
Rise time, maximum ..... Not applicable
Decay time, maximum ..... Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation ..... No failure
Aging and compatibility cable ..... No failure
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Halogen content - percent, maximum
Jacket 0.2
Fillers 0.2
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Smoke index, maximum
Jacket 25
Fillers 45
Insulation 10
Toxicity index, maximum
Jacket 5
Fillers 5
Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure
Fungus Resistance ..... No failure
Abrasion resistance ..... No failure
Gas Flame (3 hour) ..... Not applicable
Component wire ..... No failure

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Unit ordering length:	7 & 16	500 feet nominal.
	70	250 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-007)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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